

Mohammad M. T.R. Hussein

Biomedical Engineering & Bioinformatics | Computational Imaging | Digital Holographic Microscopy

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PROFILE

Biomedical engineering introduced me to programming, signal processing, electronics, medical science, CAD and simulation. Programming felt familiar because it reminded me of drawing. You begin with an empty space then build something useful from it. Curiosity led me toward computational imaging. The deeper I went, the more I enjoyed learning how light behaves and how polarization, Jones matrices and hologram reconstruction can reveal information that ordinary images miss. I also want to turn novel ideas into useful products and sustainable ventures. For me that starts with understanding the problem, studying it properly and building with purpose.

RESEARCH EXPERIENCE

2023 — PRESENT

Graduate Researcher · Optical Imaging & Polarization Analysis

Istanbul Medipol University

Developing polarization-sensitive holography systems and Python workflows for Jones matrix reconstruction, phase processing, retardance, orientation and scientific visualization.

15 JAN 2024 — PRESENT

Master’s Scholarship Researcher · Project 123N774

TÜBİTAK 2530 Türkiye–Moldova Bilateral Cooperation

Contributing to polarization-sensitive DHM with singular structured illumination for biomedical applications.

EDUCATION

15 SEP 2023 — PRESENT

MSc · Biomedical Engineering and Bioinformatics

Istanbul Medipol University

- Ongoing
- Thesis defense scheduled for 13 Aug 2026
- Focus: polarization-sensitive digital holographic microscopy

11 SEP 2019 — 10 JUN 2023

BSc · Biomedical Engineering

Istanbul Medipol University

- CGPA 3.35/4.00
- Graduated with Honors
- Ranked third in the Biomedical Engineering class according to university records

SKILLS

Computational imaging

PDHM · Jones matrix reconstruction · Polarization analysis · FFT hologram reconstruction · Phase processing · Retardance and optical-axis analysis

Programming

Python · Scientific image processing · Numerical analysis · Scientific visualization

Medical 3D & simulation

DICOM · 3D Slicer · Fusion 360 · ANSYS static structural · Finite element analysis · 3D printing

Design & product

Web design · Brand identity · Product prototyping · Research communication

SELECTED RECOGNITION

2026

Biomedical Optics Express publication

Dual-shot polarization digital holographic microscopy with phase bias equalization for full-field Jones matrix imaging.

12 SEP 2025

Third Place · Best Poster Competition · FOTONİK 2025

Enhancing Jones Matrix Reconstruction in Polarization DHM via Phase Bias Equalization · Koç University.

27–29 AUG 2025

Oral presentation · NanoTR-19

Jones Matrix Analysis via a Polarization-Sensitive Sensor in Digital Holographic Microscopy · oral format is currently supported by personal records.

SELECTED PROJECTS

Dual-shot PDHM

Full-field Jones matrix imaging with inter-shot phase bias equalization.

PDHM · Jones matrices · Python · FFT · Phase unwrapping · Polarization analysis

Reference-Free Phase Bias Equalization

An optimization-based approach that removes the need for an isotropic reference region.

Optimization · Jones matrices · Python · Phase correction · Computational imaging

Patient-Specific Knee Reconstruction & FEA

Patient-specific knee models, an implant concept and comparative structural simulations.

DICOM · 3D Slicer · Fusion 360 · ANSYS · FEA · 3D printing

VitroMech

An early biomedical 3D modeling concept focused on patient-specific anatomy.

Medical imaging · DICOM · 3D modeling · Product design · Web design

Workin

An evidence-informed, phone-first calisthenics workout tracker.

Product design · Next.js · TypeScript · Local-first data · GitHub Pages

INDEPENDENT LEARNING

- Stanford CS231N · Deep Learning for Computer Vision
- OSSU Computer Science
- The Complete Full-Stack Web Development Bootcamp
- KIT Computational Imaging